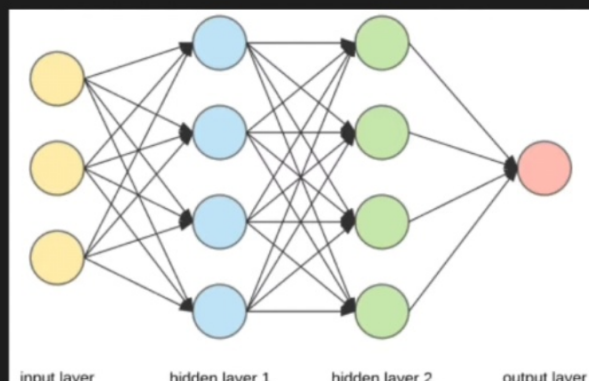
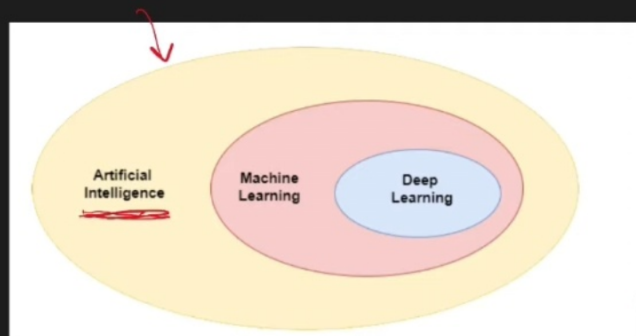


What is Deep Learning?

16 February 2022 06:32

Deep Learning is a subfield of Artificial Intelligence and Machine Learning that is inspired by the structure of a human brain.

Deep learning algorithms attempt to draw similar conclusions as humans would by continually analyzing data with a given logical structure called Neural Network.



→ ML depends on statistical technique but DL depends on logical structure called neural network and that is inspired by human brain.

→ The circles are perceptrons and the arrows are weights.

→ NN is divided by layers

input layer → hidden layer → output layer.

Deep learning is part of a broader family of machine learning methods based on artificial neural networks with representation learning.

Deep Learning Algorithms uses multiple layers to progressively extract higher-level features from the raw input. For example, in image processing, lower layers may identify edges, while higher layers may identify the concepts relevant to a human such as digits or letters or faces.

feature extraction engineering

features

Deep learning Dog / cat classifier

→ Dog

→ Cat

→ ML → features

→ size

→ color

→ Feature learning / Representation learning.

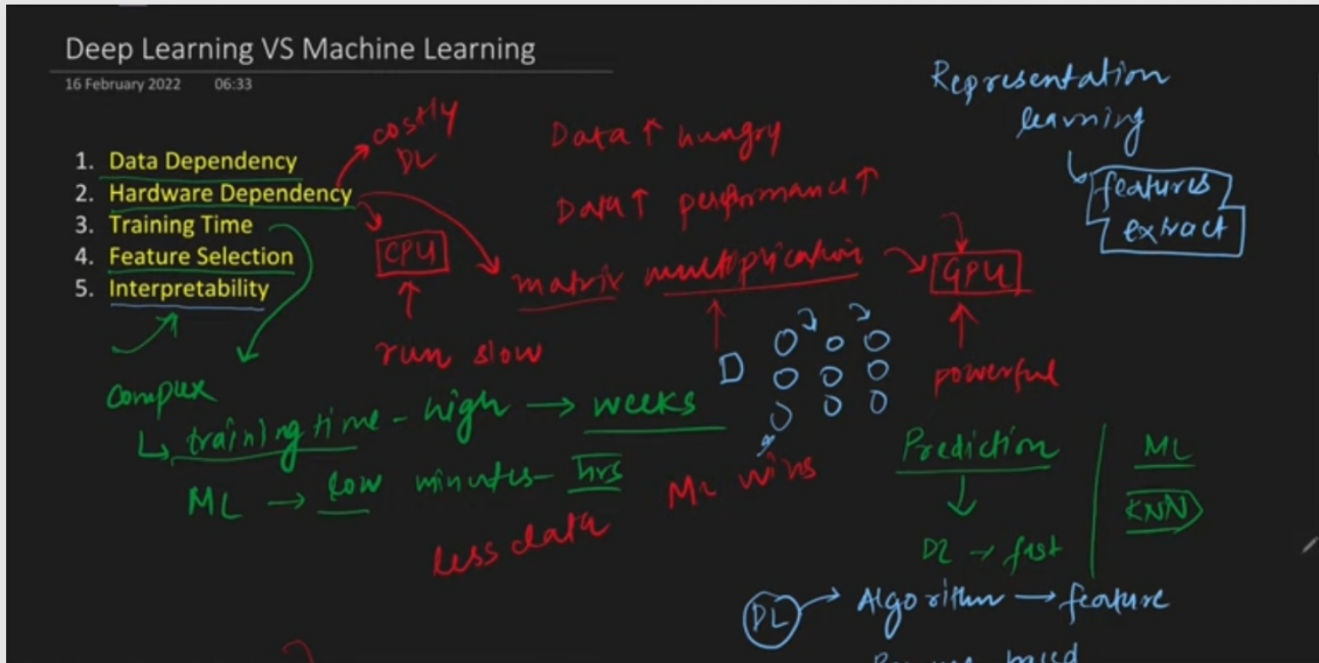
→ In ML we need to separate feature in our own and then we feed those featured data to the model.

→ In DL we do not need feature engineering.

→ In DL the first layers extract

small details where farthest layers extract detailed features.

Basic $\rightarrow$ complex.



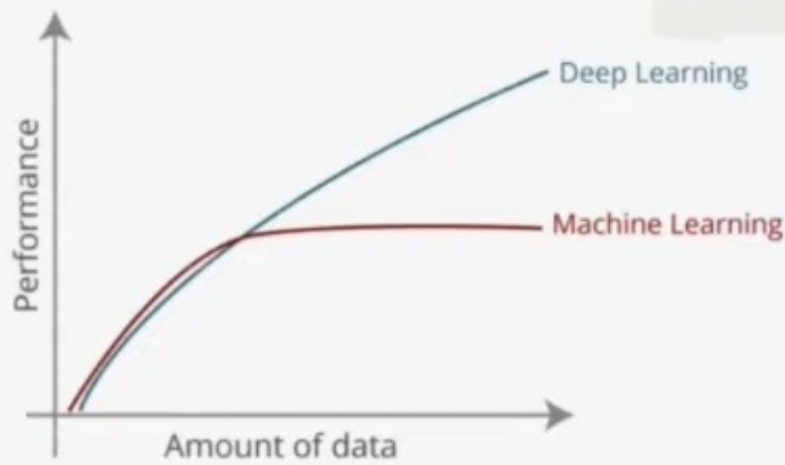
* DL VS ML

①

$\rightarrow$ DL is data hungry

$\rightarrow$ The more data it feed the more performance it provide.

$\rightarrow$ whereas ML is not good at huge data.



②

- DL need strong hardware
- Need GPU
- ML can run on CPU.

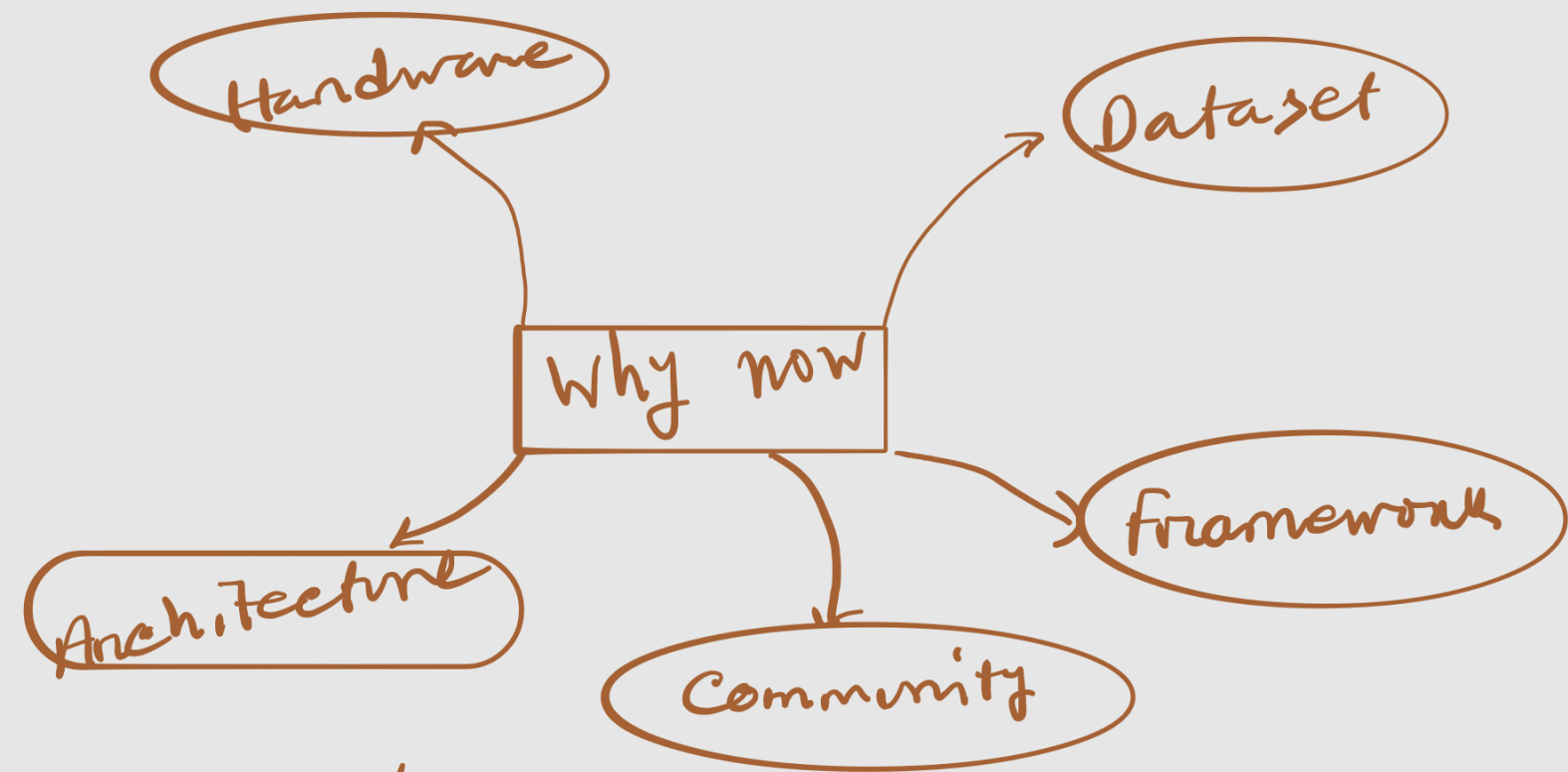
- ③ → DL needs high training time.
→ DL has fast prediction time.

④ Features selection



- ⑤ → Interpretability, DL doesn't tell us what are the features

son that it classifies classes.



① Dataset

→ Data is generating exponentially



→ Big companies publish open source labeled data so that people can research on it.

2) Hardware.

→ DL needs matrix calculation over big data.

→ To process big data faster we need high performing

hardware → GPU, TPU

3) Framework/Libraries.

→ We don't code raw to apply DL

→ Big companies develop library

so that we can use those to perform DL without writing raw code.

→ Famous library:

1) Tensorflow by Google

2) Pytorch by Facebook

→ Tensorflow was hard to use

→ Google introduce Keras to make it easy to use.

4) DL architectures

→ In DL we have network architecture.

→ For different task we need modified architecture.

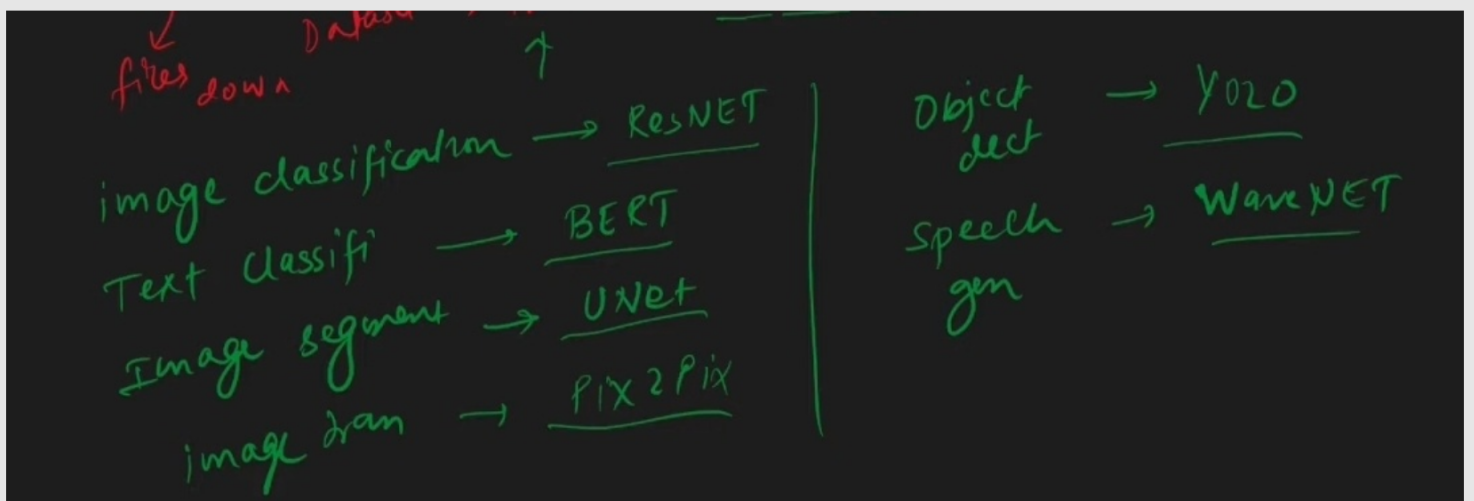
→ So, we modify architecture

and analyze which one performing well.

→ So, big companies build some arch. on specific field and trained them, so that people can use it without modify it and testing it.

→ It is called transfer learning.

Famous architectures



→ people

